Rust逆向之程序启动

本文通过对以下最简单的代码程序进行逆向，分析Rust在执行main函数之前执行的代码，来学习Rust语言。

```
fn main() {
    println!("Hello World");
}
```

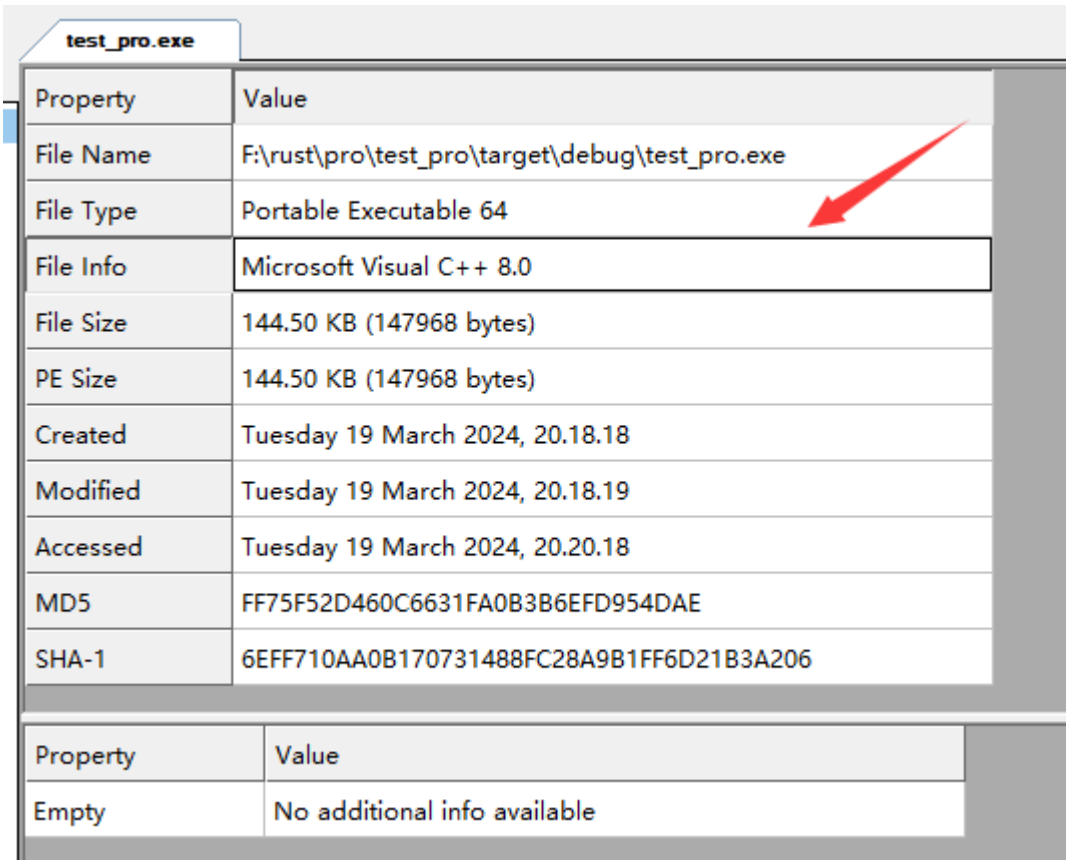
以下是编译所用的Rust版本，不知道后续更新会不会导致本文的一些信息不一样。

```
PS C:\Users\admin> rustc --version
rustc 1.76.0 (07dca489a 2024-02-04)
```

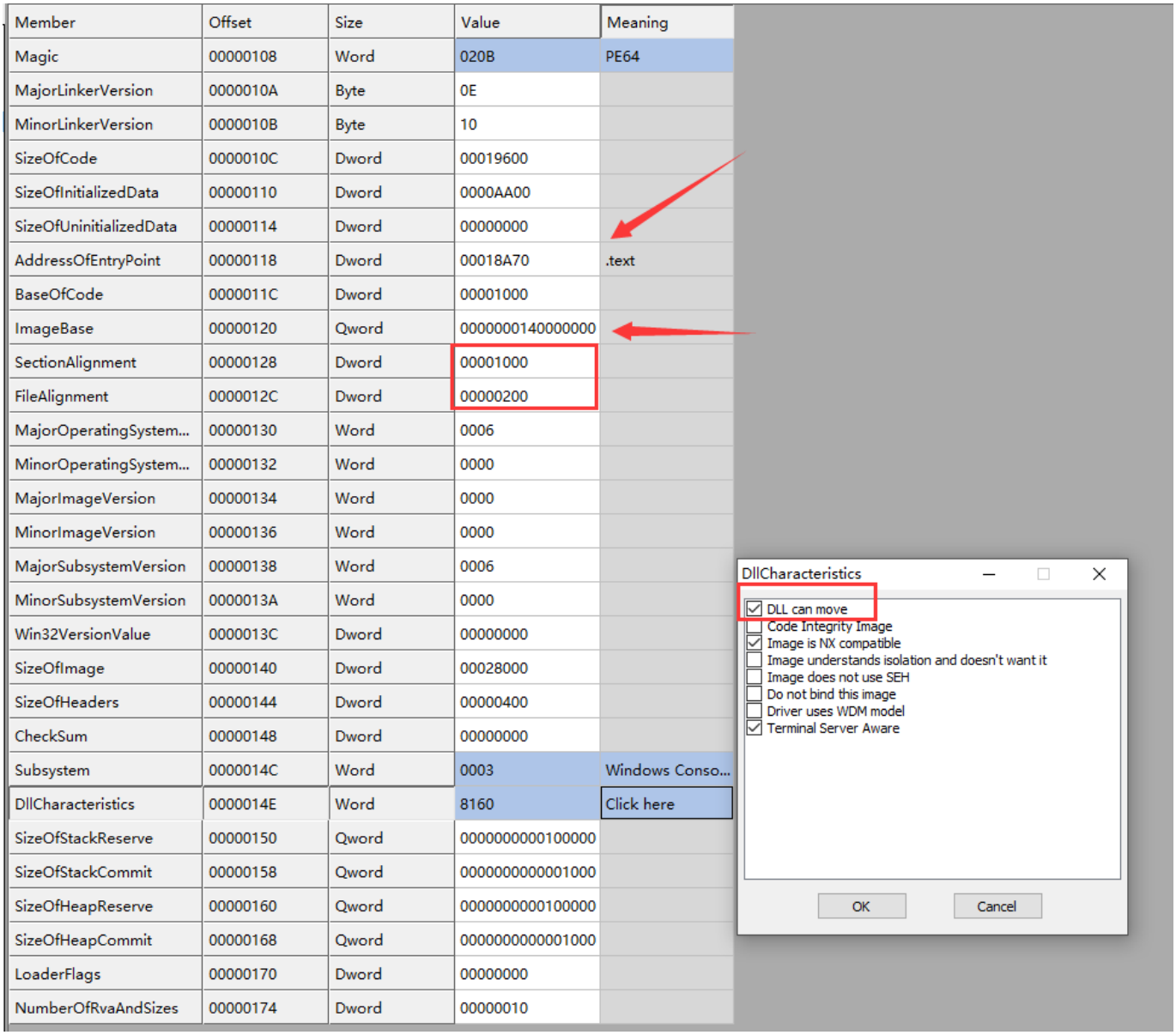
纯业余爱好，水平有限，错误难以避免，欢迎交流。

1.基本PE信息

首先可以看到VC++8.0，猜测Rust编译器可能内置了VC++8.0的编译器：



DOSHeader和FileHeader没看到有什么特别的，从OptionHeader可以知道程序的入口地址是0x140018A70。



Member	Offset	Size	Value	Meaning
Magic	00000108	Word	020B	PE64
MajorLinkerVersion	0000010A	Byte	0E	
MinorLinkerVersion	0000010B	Byte	10	
SizeOfCode	0000010C	Dword	00019600	
SizeOfInitializedData	00000110	Dword	0000AA00	
SizeOfUninitializedData	00000114	Dword	00000000	
AddressOfEntryPoint	00000118	Dword	00018A70	.text
BaseOfCode	0000011C	Dword	00001000	
ImageBase	00000120	Qword	0000000140000000	
SectionAlignment	00000128	Dword	00001000	
FileAlignment	0000012C	Dword	00000200	
MajorOperatingSystem...	00000130	Word	0006	
MinorOperatingSystem...	00000132	Word	0000	
MajorImageVersion	00000134	Word	0000	
MinorImageVersion	00000136	Word	0000	
MajorSubsystemVersion	00000138	Word	0006	
MinorSubsystemVersion	0000013A	Word	0000	
Win32VersionValue	0000013C	Dword	00000000	
SizeOfImage	00000140	Dword	00028000	
SizeOfHeaders	00000144	Dword	00000400	
CheckSum	00000148	Dword	00000000	
Subsystem	0000014C	Word	0003	Windows Conso...
DllCharacteristics	0000014E	Word	8160	Click here
SizeOfStackReserve	00000150	Qword	0000000000100000	
SizeOfStackCommit	00000158	Qword	0000000000001000	
SizeOfHeapReserve	00000160	Qword	0000000000100000	
SizeOfHeapCommit	00000168	Qword	0000000000001000	
LoaderFlags	00000170	Dword	00000000	
NumberOfRvaAndSizes	00000174	Dword	00000010	

这里比较特别的就是FileAlignment为0x200，而C/C++编译器编译出来的文件一般为0x400和0x1000，而64位的程序基本都是0x1000。这里应该是Rust做了优化，来缩小文件的大小。另外可以看到，基址随机化打开了，这里可以关掉方便后续调试。

在Data Directories这里可以看到，Rust编译出来的程序有TLS表，所以可以想到，在执行main函数之前可能执行了TLS回调函数：

Member	Offset	Size	Value	Section
Export Directory RVA	00000178	Dword	00000000	
Export Directory Size	0000017C	Dword	00000000	
Import Directory RVA	00000180	Dword	00023594	.rdata
Import Directory Size	00000184	Dword	000000B4	
Resource Directory RVA	00000188	Dword	00000000	
Resource Directory Size	0000018C	Dword	00000000	
Exception Directory RVA	00000190	Dword	00025000	.pdata
Exception Directory Size	00000194	Dword	000011F4	
Security Directory RVA	00000198	Dword	00000000	
Security Directory Size	0000019C	Dword	00000000	
Relocation Directory RVA	000001A0	Dword	00027000	.reloc
Relocation Directory Size	000001A4	Dword	000003A0	
Debug Directory RVA	000001A8	Dword	00020270	.rdata
Debug Directory Size	000001AC	Dword	00000054	
Architecture Directory RVA	000001B0	Dword	00000000	
Architecture Directory Size	000001B4	Dword	00000000	
Reserved	000001B8	Dword	00000000	
Reserved	000001BC	Dword	00000000	
TLS Directory RVA	000001C0	Dword	000203D0	.rdata
TLS Directory Size	000001C4	Dword	00000028	
Configuration Directory RVA	000001C8	Dword	000202D0	.rdata
Configuration Directory Size	000001CC	Dword	00000100	
Bound Import Directory RVA	000001D0	Dword	00000000	
Bound Import Directory Size	000001D4	Dword	00000000	
Import Address Table Direct...	000001D8	Dword	0001B000	.rdata
Import Address Table Direct...	000001DC	Dword	000002A0	
Delay Import Directory RVA	000001E0	Dword	00000000	
Delay Import Directory Size	000001E4	Dword	00000000	
.NET MetaData Directory RVA	000001E8	Dword	00000000	
.NET MetaData Directory Size	000001EC	Dword	00000000	

节区这里没什么特别的，这里看出Rust不像Go会增加什么额外的节区：

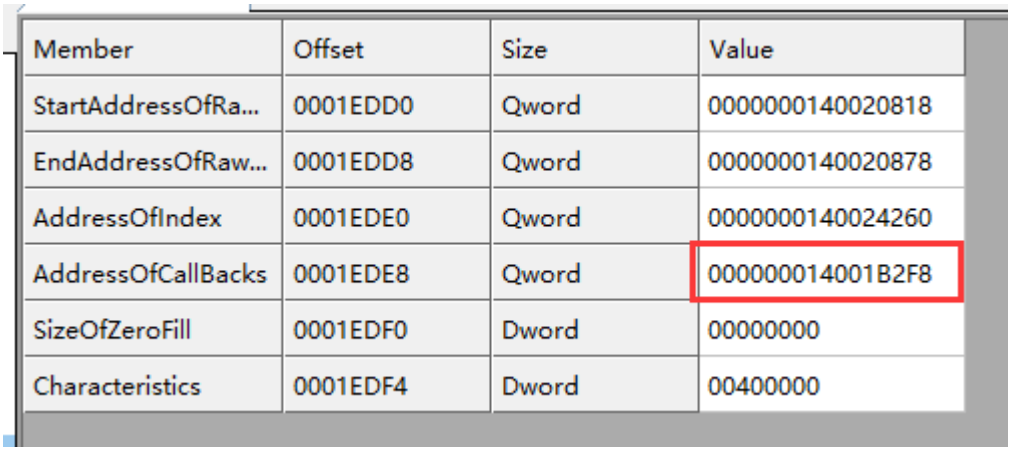
Name	Virtual Size	Virtual Addr...	Raw Size	Raw Address	Reloc Address	Linenumbers	Relocations N...	Linenumbers ...	Characteristics
Byte[8]	Dword	Dword	Dword	Dword	Dword	Dword	Word	Word	Dword
.text	0001955A	00001000	00019600	00000400	00000000	00000000	0000	0000	60000020
.rdata	00008F90	0001B000	00009000	00019A00	00000000	00000000	0000	0000	40000040
.data	000002E8	00024000	00000200	00022A00	00000000	00000000	0000	0000	C0000040
.pdata	000011F4	00025000	00001200	00022C00	00000000	00000000	0000	0000	40000040
.reloc	000003A0	00027000	00000400	00023E00	00000000	00000000	0000	0000	42000040

而导入表这里可以看到，Rust程序会导入一些Rust专有的Dll和函数：

Module Name	Imports	OFTs	TimeDateSta...	ForwarderCha...	Name RVA	FTs (IAT)
00022856	N/A	00021FD0	00021FD4	00021FD8	00021FDC	00021FE0
szAnsi	(nFunctions)	Dword	Dword	Dword	Dword	Dword
KERNEL32.dll	43	00023648	00000000	00000000	00023BE0	0001B000
ntdll.dll	2	000238D0	00000000	00000000	00023C14	0001B288
VCRUNTIME140.dll	7	000237A8	00000000	00000000	00023C8A	0001B160
api-ms-win-crt-runtime-l1-1-0.dll	18	00023820	00000000	00000000	00023E56	0001B1D8
api-ms-win-crt-math-l1-1-0.dll	1	00023810	00000000	00000000	00023E78	0001B1C8
api-ms-win-crt-stdio-l1-1-0.dll	2	000238B8	00000000	00000000	00023E98	0001B270
api-ms-win-crt-locale-l1-1-0.dll	1	00023800	00000000	00000000	00023EB8	0001B1B8
api-ms-win-crt-heap-l1-1-0.dll	2	000237E8	00000000	00000000	00023EDA	0001B1A0

OFTs	FTs (IAT)	Hint	Name
Qword	Qword	Word	szAnsi
0000000000023D...	0000000000023D...	0036	_initterm
0000000000023D...	0000000000023D...	0037	_initterm_e
0000000000023D...	0000000000023D...	0055	exit
0000000000023D...	0000000000023D...	0023	_exit
0000000000023D...	0000000000023D...	0028	_get_initial_narrow_environment
0000000000023D...	0000000000023D...	0004	__p__argc
0000000000023D...	0000000000023D...	0005	__p__argv
0000000000023D...	0000000000023D...	0016	_cexit
0000000000023D...	0000000000023D...	0015	_c_exit
0000000000023D...	0000000000023D...	003D	_register_thread_local_exe_atexi...
0000000000023C...	0000000000023C...	0018	_configure_narrow_argv
0000000000023C...	0000000000023C...	0040	_seh_filter_exe
0000000000023C...	0000000000023C...	0033	_initialize_narrow_environment

以下是TLS表信息，这里可以看到保存TLS回调函数的地址：



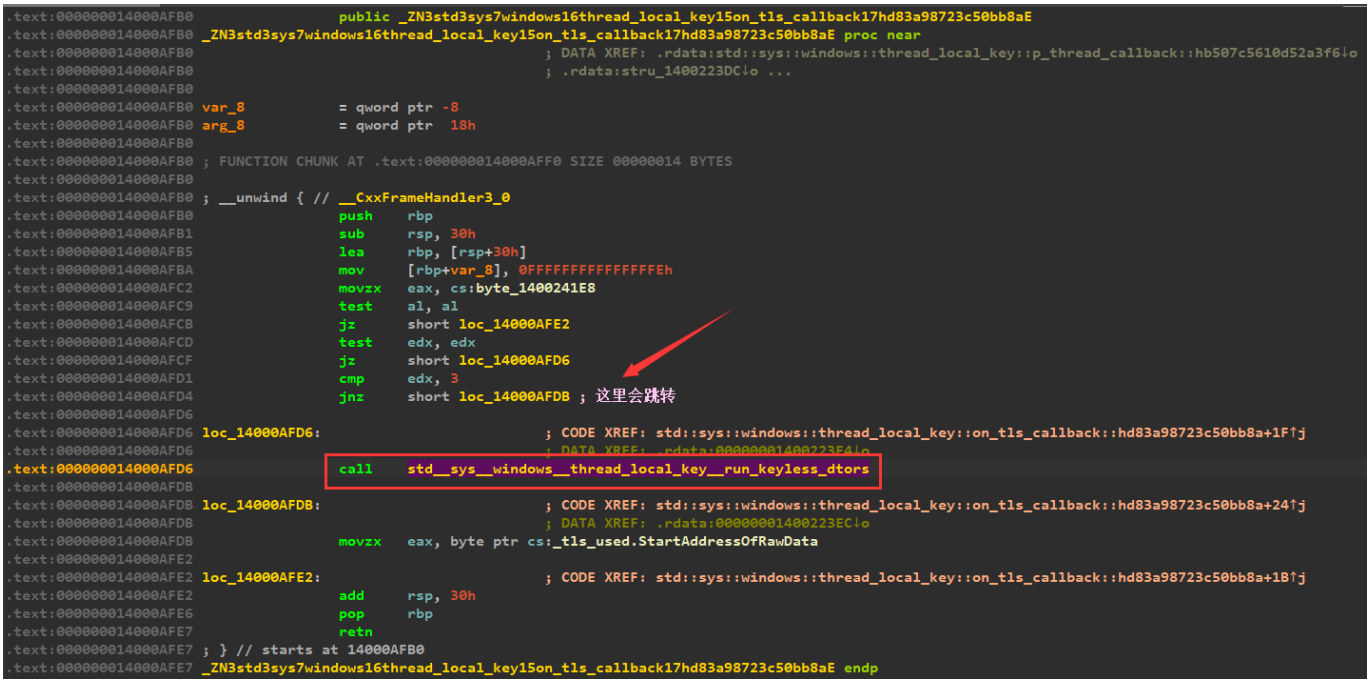
Member	Offset	Size	Value
StartAddressOfRa...	0001EDD0	Qword	0000000140020818
EndAddressOfRaw...	0001EDD8	Qword	0000000140020878
AddressOfIndex	0001EDE0	Qword	0000000140024260
AddressOfCallBacks	0001EDE8	Qword	000000014001B2F8
SizeOfZeroFill	0001EDF0	Dword	00000000
Characteristics	0001EDF4	Dword	00400000

2.静态调试

根据保存TLS回调函数地址，可以找到要执行的TLS回调函数，这里可以看到只需要执行_ZN3std3sys7windows16thread_local_key15on_tls_callback17hd83a98723c50bb8aE这个回调函数：

```
.rdata:000000014001B2F8 ;
std::sys::windows::thread_local_key::p_thread_callback::hb507c5610d52a3f6
.rdata:000000014001B2F8
_ZN3std3sys7windows16thread_local_key17p_thread_callback17hb507c5610d52a3f6E dq offset
_ZN3std3sys7windows16thread_local_key15on_tls_callback17hd83a98723c50bb8aE ;
std::sys::windows::thread_local_key::on_tls_callback::hd83a98723c50bb8a
```

这个回调函数会在经过一些判断以后，决定是否执行std__sys__windows__thread_local_key__run_keyless_dtors这个函数。但是经过动态调试，发现会在上一个判断跳转。也就是说第二个参数不等于3，导致跳转，不执行这个函数。



接下来就会执行入口函数mainCRTStartup，这里会首先调用__security_init_cookie函数，这个函数会根据时间，进程PID和线程PID来设置cookie，随后跳转到__scrt_common_main_seh函数执行：

```
.text:0000000140018A70 ; int __fastcall mainCRTStartup()
.text:0000000140018A70                 public mainCRTStartup
.text:0000000140018A70 mainCRTStartup  proc near               ; DATA XREF:
.pdata:0000000140025F6C↓o
.text:0000000140018A70                                         ;
.pdata:0000000140025F78↓o
.text:0000000140018A70                 sub     rsp, 28h
.text:0000000140018A74                 call    __security_init_cookie ; 设置cookie
.text:0000000140018A79                 add     rsp, 28h
.text:0000000140018A7D                 jmp     __scrt_common_main_seh
.text:0000000140018A7D mainCRTStartup  endp
```

__scrt_common_main_seh函数，会首先调用一些Rust自定义的函数(包括上面说的自定义的.dll中的函数)进行一些设置。随后获取argc和argv作为参数来调用main函数：

```
.text:00000001400189DE                 call    __p___argv_0
.text:00000001400189E3                 mov     rdi, [rax]
.text:00000001400189E6                 call    __p___argc_0
.text:00000001400189EB                 mov     rbx, rax
.text:00000001400189EE                 call    _get_initial_narrow_environment_0
.text:00000001400189F3                 mov     r8, rax         ; envp
.text:00000001400189F6                 mov     rdx, rdi        ; argv
.text:00000001400189F9                 mov     ecx, [rbx]      ; argc
.text:00000001400189FB                 call    main
.text:0000000140018A00                 mov     ebx, eax
```

main函数会调用std::rt::lang_start::h09171374b7c116e9函数，其中第一个参数就是编写Rust函数的时候写的fn main()函数的地址。从这可以看出Rust将它重命名为"文件名__main"这样一个函数名。然后第二个和第三个参数则分别是argc和argv，最后第四个参数则为0：

```
.text:0000000140001240 ; int __fastcall main(int argc, const char **argv, const char
**envp)
.text:0000000140001240 main            proc near               ; CODE XREF:
__scrt_common_main_seh+107↓p
.text:0000000140001240                                         ; DATA XREF:
.pdata:0000000140025084↓o
.text:0000000140001240                 sub     rsp, 28h
.text:0000000140001244                 mov     r8, rdx         ; unsigned __int8 **
.text:0000000140001247                 movsxd  rdx, ecx        ; __int64
.text:000000014000124A                 lea     rcx, test_pro__main ; void (__fastcall
*)()
.text:0000000140001251                 xor     r9d, r9d        ; unsigned __int8
.text:0000000140001254                 call
_ZN3std2rt10lang_start17h09171374b7c116e9E ; std::rt::lang_start::h09171374b7c116e9
.text:0000000140001259                 nop
.text:000000014000125A                 add     rsp, 28h
.text:000000014000125E                 retn
.text:000000014000125E main            endp
```

std::rt::lang_start::h09171374b7c116e9函数则会将参数赋值到args数组中。然后将其地址作为第一个参数，并将impl__std::rt::lang_start::closure_env_0_tuple_____core::ops::function::Fn_tuple_____::vtable_数组地址作为第二个参数，调用std::rt::lang_start_internal::h68f55995b3e4e811函数。这里不知道为什么test_pro__main函数地址会被赋值两次，不过这里只要记得args数组第一个元素是test_pro__main函数地址就行。

```
__int64 __fastcall std::rt::lang_start::h09171374b7c116e9(
        void (__fastcall *test_pro__main)(),
        __int64 arg0,
        unsigned __int8 **argv,
        byte a4)
{
  byte args[40]; // [rsp+38h] [rbp-30h] BYREF

  *(_QWORD *)&args[8] = test_pro__main;
  *(_QWORD *)&args[0x10] = arg0;
  *(_QWORD *)&args[0x18] = argv;
  args[0x27] = a4;
  *(_QWORD *)args = test_pro__main;
  return std::rt::lang_start_internal::h68f55995b3e4e811(
           (__int64)args,

(__int64)&impl__std::rt::lang_start::closure_env_0_tuple_______core::ops::function::Fn
_tuple_______::vtable_);
}
```

impl__std::rt::lang_start::closure_env_0_tuple_____core::ops::function::Fn_tuple_____::vtable_数组中保存了一些函数的地址，这里需要注意最后一个函数，因为后面会执行它：

```
.rdata:000000014001B3B8 ; impl$<std::rt::lang_start::closure_env$0<tuple$<> >,
core::ops::function::Fn<tuple$<> > >::vtable_type$
impl__std::rt::lang_start::closure_env_0_tuple_______core::ops::function::Fn_tuple____
___::vtable_
.rdata:000000014001B3B8
impl$_std__rt__lang_start__closure_env$0_tuple$_____core__ops__function__Fn_tuple$___
_____vtable$ impl$<std::rt::lang_start::closure_env$0<tuple$<> >,
core::ops::function::Fn<tuple$<> > >::vtable_type$ <offset
_ZN4core3ptr85drop_in_place$LT$std__rt__lang_start$LT$$LP$$RP$$GT$__$u7b$$u7b$closure$
u7d$$u7d$$GT$17hc6641ac3297587c8E,\
.rdata:000000014001B3B8                                         ; DATA XREF:
std::rt::lang_start::h09171374b7c116e9+2A↑o
.rdata:000000014001B3B8
8,\ ;
core::ptr::drop_in_place$LT$std..rt..lang_start$LT$$LP$$RP$$GT$..$u7b$$u7b$closure$u7d
$$u7d$$GT$::hc6641ac3297587c8
.rdata:000000014001B3B8
8,\
.rdata:000000014001B3B8
offset
_ZN4core3ops8function6FnOnce40call_once$u7b$$u7b$vtable_shim$u7d$$u7d$17h60ff440d68689
0ddE,\
.rdata:000000014001B3B8
offset _ZN3std2rt10lang_start28_$u7b$$u7b$closure$u7d$$u7d$17h37c065078fbc660eE,\
.rdata:000000014001B3B8
offset _ZN3std2rt10lang_start28_$u7b$$u7b$closure$u7d$$u7d$17h37c065078fbc660eE>
```

std::rt::lang_start_internal::h68f55995b3e4e811函数首先会将两个参数保存在局部变量var_40和var_48中。然后将std__sys__windows__stack_overflow__vectored_handler增到SEH链表中，该函数用于处理栈溢出异常的情况。接着会设置最大堆栈大小。如果上述两个函数执行失败，则会跳转到报错代码执行：

```
.text:0000000140003090 ; std::rt::lang_start_internal::h68f55995b3e4e811
.text:0000000140003090 _ZN3std2rt19lang_start_internal17h68f55995b3e4e811E proc near
.text:0000000140003090                                         ; CODE XREF:
std::rt::lang_start::h09171374b7c116e9+35↑p
.text:0000000140003090                                         ; DATA XREF:
.rdata:stru_140020EA8↓o ...
.text:0000000140003090 StackSizeInBytes= dword ptr -98h
.text:0000000140003090 var_48          = qword ptr -48h
.text:0000000140003090 var_40          = qword ptr -40h
.text:0000000140003090                 push    rbp
.text:0000000140003091                 push    rsi
.text:0000000140003092                 sub     rsp, 0F8h
.text:0000000140003099                 lea     rbp, [rsp+80h]
.text:00000001400030A1                 mov     [rbp+80h+var_30], 0FFFFFFFFFFFFFFFEh
.text:00000001400030A9                 mov     [rbp+80h+var_40], rdx ; 这里保存了第二个
参数，即
impl__std::rt::lang_start::closure_env_0_tuple_____core::ops::function::Fn_tuple____
___::vtable_数组地址
.text:00000001400030AD                 mov     [rbp+80h+var_48], rcx ; 这里保存了第一个
参数，即args的地址
.text:00000001400030B1                 lea     rdx,
std__sys__windows__stack_overflow__vectored_handler ; Handler
.text:00000001400030B8                 xor     ecx, ecx        ; First
.text:00000001400030BA                 call    cs:__imp_AddVectoredExceptionHandler ;
增加SEH
.text:00000001400030C0                 test    rax, rax
.text:00000001400030C3                 jz      loc_1400031A4
.text:00000001400030C9                 mov     [rbp+80h+StackSizeInBytes], 5000h
.text:00000001400030D0                 lea     rcx, [rbp+80h+StackSizeInBytes] ;
StackSizeInBytes
.text:00000001400030D4                 call    cs:__imp_SetThreadStackGuarantee ; 设置
堆栈最大大小
.text:00000001400030DA                 test    eax, eax
.text:00000001400030DC                 jnz     short loc_1400030ED ; 设置成功则跳转
.text:00000001400030DE                 call    cs:__imp_GetLastError
.text:00000001400030E4                 cmp     eax, 78h ; 'x'  ; 这个错误码表示该系统不
支持这个功能
.text:00000001400030E7                 jnz     loc_140003267
```

如果执行成功，接下来会调用Rust的一些库函数继续进行一些设置：

```
.text:00000001400030ED loc_1400030ED:                          ; CODE XREF:
std::rt::lang_start_internal::h68f55995b3e4e811+4C↑j
.text:00000001400030ED                                         ; DATA XREF:
.rdata:0000000140020EB0↓o
.text:00000001400030ED ;   try {
.text:00000001400030ED                 lea     rcx, aMain      ; "main"
.text:00000001400030F4                 mov     edx, 5
.text:00000001400030F9                 call
_ZN3std3sys7windows6thread6Thread8set_name17h4b2931aba1212a8aE ;
std::sys::windows::thread::Thread::set_name::h4b2931aba1212a8a
.text:00000001400030FE                 lea     rdx, unk_14001BA30
.text:0000000140003105                 lea     rsi, [rbp+80h+var_B8]
.text:0000000140003109                 mov     r8d, 4
.text:000000014000310F                 mov     rcx, rsi
.text:0000000140003112                 call
_ZN72_$LT$$RF$str$u20$as$u20$alloc__ffi__c_str__CString__new__SpecNewImpl$GT$13spec_ne
w_impl17h655a14117da83755E ;
_$LT$$RF$str$u20$as$u20$alloc..ffi..c_str..CString..new..SpecNewImpl$GT$::spec_new_imp
l::h655a14117da83755
.text:0000000140003117                 mov     rax, [rbp+80h+var_B8]
.text:000000014000311B                 mov     rcx, 8000000000000000h
.text:0000000140003125                 mov     [rbp+80h+var_50], rax
.text:0000000140003129                 cmp     rax, rcx
.text:000000014000312C                 jnz     loc_1400031DE
.text:0000000140003132                 mov     rcx, [rbp+80h+var_B0]
.text:0000000140003136                 mov     rdx, [rbp+80h+var_A8]
.text:0000000140003136 ;   } // starts at 1400030ED
.text:000000014000313A
.text:000000014000313A loc_14000313A:                          ; DATA XREF:
.rdata:0000000140020EB8↓o
.text:000000014000313A ;   try {
.text:000000014000313A                 call
_ZN3std6thread6Thread3new17h8c9c8375c980fc00E ;
std::thread::Thread::new::h8c9c8375c980fc00
.text:000000014000313A ;   } // starts at 14000313A
.text:000000014000313F
.text:000000014000313F loc_14000313F:                          ; DATA XREF:
.rdata:0000000140020EC0↓o
.text:000000014000313F ;   try {
.text:000000014000313F                 mov     rcx, rax
.text:0000000140003142                 call
_ZN3std10sys_common11thread_info3set17hb9b8f494a7e09866E ;
std::sys_common::thread_info::set::hb9b8f494a7e09866
```

接下来，函数会将在上面保存的第一个参数地址赋值给rcx，第二个参数地址付给rax。此时rax保存的就是impl__std::rt::lang_start::closure_env_0_tuple_____core::ops::function::Fn_tuple_____::vtable_数组地址。随后，将会调用该地址偏移0x28处保存的函数。根据上面展示的数组内容可以知道，这

个时候会调用_ZN3std2rt10lang_start28_$u7b$u7bclosure$u7d$u7d17h37c065078fbc660eE这个函数。而调用这个函数的时候，第一个参数就是前面传过来的args数组的地址。

```
.text:0000000140003147 loc_140003147:                          ; CODE XREF:
std::rt::lang_start_internal::h68f55995b3e4e811+290↓j
.text:0000000140003147                                         ; DATA XREF:
.rdata:0000000140020EC8↓o
.text:0000000140003147                 mov     rcx, [rbp+80h+var_48]   ; 将第一个参数赋
值给r
.text:000000014000314B                 mov     rax, [rbp+80h+var_40] ; 将第二个参数赋值
给rax
.text:000000014000314F                 call    qword ptr [rax+28h]
```

*ZN3std2rt10lang_start28*$u7b$$u7b$closure$u7d$$u7d$17h37c065078fbc660eE函数会从args数组中取出第一个元素，也就是test_pro__main函数地址作为参数，接着调用std::sys_common::backtrace::__rust_begin_short_backtrace::h3cd2d30df04669c3函数：

```
.text:00000001400011E0 ; int __fastcall
std::rt::lang_start::_$u7b$$u7b$closure$u7d$$u7d$::h37c065078fbc660e(std::rt::lang_sta
rt::closure_env$0<tuple$<> > *)
.text:00000001400011E0
_ZN3std2rt10lang_start28_$u7b$$u7b$closure$u7d$$u7d$17h37c065078fbc660eE proc near
.text:00000001400011E0                                          ; CODE XREF:
core__ops__function__FnOnce__call_once_std__rt__lang_start__closure_env$0_tuple$_____t
uple$____+1A↑p
.text:00000001400011E0                                          ; DATA XREF:
.rdata:impl$_std__rt__lang_start__closure_env$0_tuple$_____core__ops__function__Fn_tu
ple$________vtable$↓o ...
.text:00000001400011E0
.text:00000001400011E0 var_14          = dword ptr -14h
.text:00000001400011E0 var_10          = qword ptr -10h
.text:00000001400011E0 var_8           = qword ptr -8
.text:00000001400011E0
.text:00000001400011E0                 sub     rsp, 38h
.text:00000001400011E4                 mov     [rsp+38h+var_10], rcx
.text:00000001400011E9                 mov     rcx, [rcx]      ; rcx=test_pro__main函数
地址
.text:00000001400011EC                 call
_ZN3std10sys_common9backtrace28__rust_begin_short_backtrace17h3cd2d30df04669c3E ;
std::sys_common::backtrace::__rust_begin_short_backtrace::h3cd2d30df04669c3
.text:00000001400011F1                 call
_ZN54_$LT$$LP$$RP$$u20$as$u20$std__process__Termination$GT$6report17h898d9242579cf8b8E
; _$LT$$LP$$RP$$u20$as$u20$std..process..Termination$GT$::report::h898d9242579cf8b8
.text:00000001400011F6                 mov     [rsp+38h+var_14], eax
.text:00000001400011FA                 lea     rax, [rsp+38h+var_14]
.text:00000001400011FF                 mov     [rsp+38h+var_8], rax
.text:0000000140001204                 mov     eax, [rsp+38h+var_14]
.text:0000000140001208                 add     rsp, 38h
.text:000000014000120C                 retn
.text:000000014000120C
_ZN3std2rt10lang_start28_$u7b$$u7b$closure$u7d$$u7d$17h37c065078fbc660eE endp
```

std::sys_common::backtrace::__rust_begin_short_backtrace::h3cd2d30df04669c3函数会继续调用core::ops::function::FnOnce::call_once::h7a03834函数:

```
.text:0000000140001170 ; void __fastcall
std::sys_common::backtrace::__rust_begin_short_backtrace::h3cd2d30df04669c3(void
(__fastcall *)())
.text:0000000140001170
_ZN3std10sys_common9backtrace28__rust_begin_short_backtrace17h3cd2d30df04669c3E proc
near
.text:0000000140001170                                          ; CODE XREF:
std::rt::lang_start::_$u7b$$u7b$closure$u7d$$u7d$::h37c065078fbc660e+C↓p
.text:0000000140001170                                          ; DATA XREF:
.pdata:0000000140025054↓o
.text:0000000140001170
.text:0000000140001170 var_8           = qword ptr -8
.text:0000000140001170
.text:0000000140001170                 sub     rsp, 38h
.text:0000000140001174                 mov     [rsp+38h+var_8], rcx
.text:0000000140001179                 call
_ZN4core3ops8function6FnOnce9call_once17h7a0383449d9a25ccE ;
core::ops::function::FnOnce::call_once::h7a0383449d9a25cc
.text:000000014000117E                 nop
.text:000000014000117F                 add     rsp, 38h
.text:0000000140001183                 retn
.text:0000000140001183
_ZN3std10sys_common9backtrace28__rust_begin_short_backtrace17h3cd2d30df04669c3E endp
```

在a25ccE ; core::ops::function::FnOnce::call_once::h7a03834函数中，才调用了test_pro__main函数

```
.text:0000000140001020 ; void __fastcall
core::ops::function::FnOnce::call_once::h7a0383449d9a25cc(void (__fastcall *)())
.text:0000000140001020 _ZN4core3ops8function6FnOnce9call_once17h7a0383449d9a25ccE proc
near
.text:0000000140001020                                          ; CODE XREF:
std::sys_common::backtrace::__rust_begin_short_backtrace::h3cd2d30df04669c3+9↓p
.text:0000000140001020                                          ; DATA XREF:
.pdata:000000014002500C↓o
.text:0000000140001020
.text:0000000140001020 var_8           = qword ptr -8
.text:0000000140001020
.text:0000000140001020                 sub     rsp, 38h
.text:0000000140001024                 mov     [rsp+38h+var_8], rcx
.text:0000000140001029                 call    rcx             ; 调用test_pro__main函数
.text:000000014000102B                 nop
.text:000000014000102C                 add     rsp, 38h
.text:0000000140001030                 retn
.text:0000000140001030 _ZN4core3ops8function6FnOnce9call_once17h7a0383449d9a25ccE endp
```

test_pro__main就是编写的输出Hello World的代码:

```
.text:0000000140001210 ; void __fastcall test_pro::main()
.text:0000000140001210 test_pro__main  proc near               ; DATA XREF: main+A↓o
.text:0000000140001210                                         ;
.pdata:0000000140025078↓o
.text:0000000140001210
.text:0000000140001210 var_30          = byte ptr -30h
.text:0000000140001210
.text:0000000140001210                 sub     rsp, 58h
.text:0000000140001214                 lea     rcx, [rsp+58h+var_30]
.text:0000000140001219                 lea     rdx, off_14001B3F8 ; "Hello World\n"
.text:0000000140001220                 mov     r8d, 1
.text:0000000140001226                 call
_ZN4core3fmt9Arguments9new_const17hcc6d1cc21b852a27E ;
core::fmt::Arguments::new_const::hcc6d1cc21b852a27
.text:000000014000122B                 lea     rcx, [rsp+58h+var_30]
.text:0000000140001230                 call
_ZN3std2io5stdio6_print17h809a7cb8277a4b47E ;
std::io::stdio::_print::h809a7cb8277a4b47
.text:0000000140001235                 nop
.text:0000000140001236                 add     rsp, 58h
.text:000000014000123A                 retn
.text:000000014000123A test_pro__main  endp
```

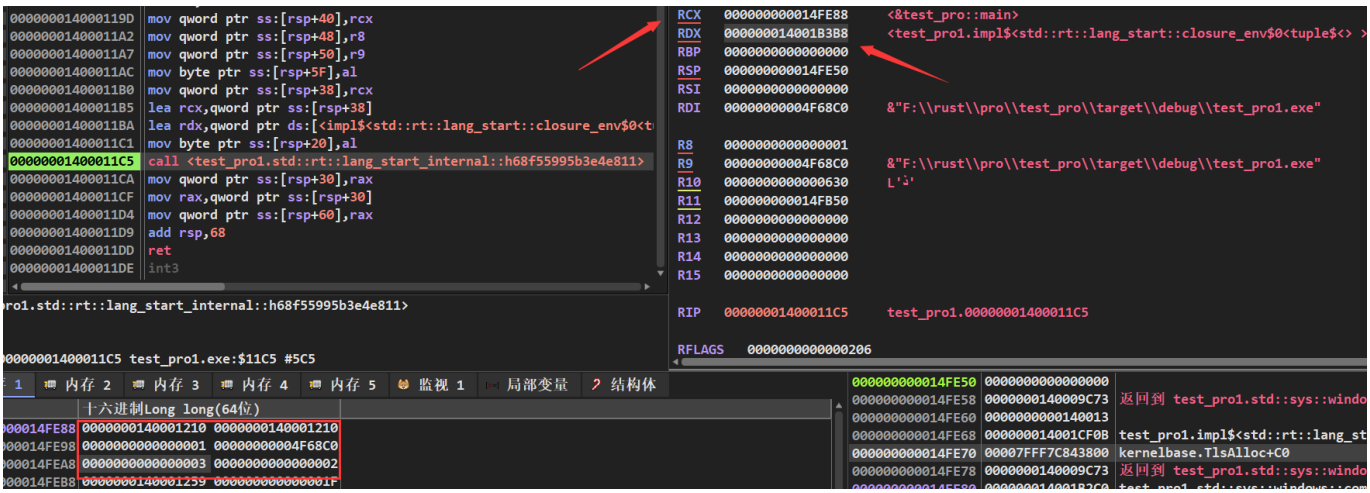
3.动态调试

接下来通过动态调试验证一下上面的一些关键步骤。首先是在调用main函数，这里可以看到前两个参数分别被赋予了argc和argv，而第三个参数则指向公共文件夹。

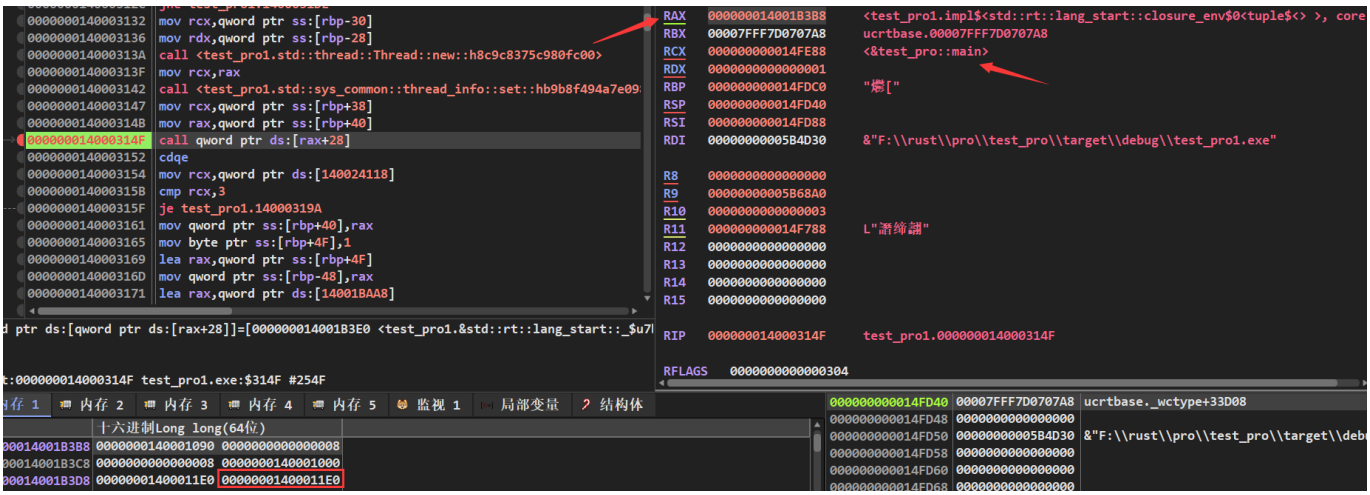
在main函数中，会将编写Rust时候的main，也就是静态分析时候说的test_pro__main作为第一个参数。argc和argv依次作为第二和第三个参数，最后一个参数则为0，随后调用std::rt::lang_start::h09171374b7c116e9函数。

std::rt::lang_start::h09171374b7c116e9函数会继续调用std::rt::lang_start_internal::h68f55995b3e4e811函数。此时，第一个参数rcx指向了局部变量数

组，数组中前两个元素保存了test_pro__main函数地址，后面依次是之前传进来的三个参数。第二个参数rdx则是impl__std::rt::lang_start::closure_env_0_tuple_____core::ops::function::Fn_tuple_____::vtable_数组地址。

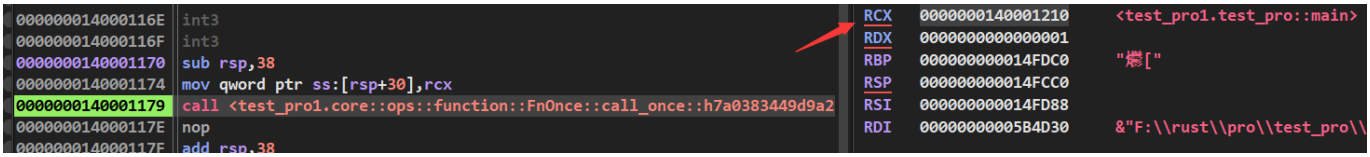


继续跟进std::rt::lang_start_internal::h68f55995b3e4e811函数，可以看到在此处rax保存的就是impl__std::rt::lang_start::closure_env_0_tuple_____core::ops::function::Fn_tuple_____::vtable_数组地址。接下来会调用这个数组中的第五个元素保存的函数，这里可以看到函数地址为0x1400011E0。而第一个参数，和上面一样是那个保存了test_pro__main函数地址的局部变量数组。

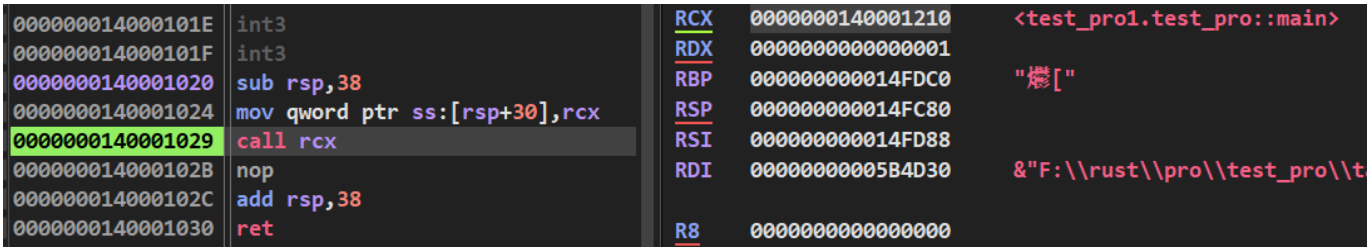


继续跟进，接下来会将test_pro__main函数作为第一个参数，然后调用std::sys_common::backtrace::__rust_begin_short_backtrace::h3cd2d30df04669c3函数。

同样的，接下来会将test_pro__main作为第一个参数，继续调用core::ops::function::FnOnce::call_once::h7a03834函数。



在core::ops::function::FnOnce::call_once::h7a03834函数，才调用test_pro__main函数。



接下来才开始正式执行fn main函数的代码：